

Prathamesh Anvekar

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EDUCATION

Vishwakarma Institute of Technology, Pune 2023 - 2027
B.Tech - Electronics and Telecommunication Engineering CGPA: 8.98

WORK EXPERIENCE

Qriocity Jan 2026 - Present
Machine Learning Developer

- Designed and deployed end-to-end AI/ML systems across domains including **medical imaging, EEG-based sleep stage classification, waste detection, and automated disease diagnosis pipelines.**
- Built and optimized deep learning architectures using **PyTorch and TensorFlow**, implementing **CNNs for image tasks, BiLSTMs for temporal modeling, GANs for data augmentation, and Transformer-based models for sequence understanding.**
- Engineered robust data preprocessing and augmentation pipelines, performed **feature engineering, hyperparameter tuning**, and systematic model evaluation to ensure production-grade accuracy and reliability.
- Deployed scalable inference APIs using FastAPI**, integrated with React-based frontends, enabling real-world usability and low-latency model serving.

PROJECTS

Linux Process Isolation Engine (seccomp-BPF, Kernel-Enforced) 2026
Systems Programming | C, Linux, libseccomp, OS Internals *GitHub*

- Built a **user-space sandboxing engine** enforcing fine-grained syscall restrictions using **Linux seccomp-BPF filters.**
- Designed and compiled **policy-driven syscall filters** to restrict process behavior including file access, networking, and process creation (fork/clone).
- Implemented a **secure process execution pipeline** using fork/exec with kernel-level enforcement of constraints.
- Developed a **syscall auditing and logging system** for runtime violation tracing and debugging.
- Benchmarked sandboxed vs native execution to analyze **performance overhead and trade-offs.**

Secure LLM Proxy Server 2026
Secure LLM Infrastructure | FastAPI, Quantization, RAG *GitHub*

- Architected a security-first LLM gateway** using **FastAPI** middleware to sanitize PII, enforce input validation, and mitigate prompt injection and jailbreak attacks.
- Integrated **Google Gemma 3 with 4-bit quantization (BitsAndBytes) and LoRA adapters** to enable efficient local inference while reducing memory footprint.
- Designed a **multi-stage threat detection pipeline** combining regex-based filters and ML classifiers to secure RAG workflows and external web queries via **SerpAPI.**
- Developed a **React and Tailwind dashboard** to visualize real-time security telemetry, blocked requests, and system activity logs.

AWARDS & ACHIEVEMENTS

- Top 150 – Amazon ML Challenge 2025:** Ranked among **top 150 out of 80,000+ participants** nationwide in a competitive machine learning challenge.
- Top 4 – WOW Verse Hackathon (Google GDG Sponsored) 2025:** Selected among top 4 teams and invited to present the solution to Google Developers through the GDG program.

TECHNICAL SKILLS

- Programming:** C, C++, Python, Java, Go, JavaScript
- Backend & Frameworks:** FastAPI, Node.js, Express.js, ReactJS
- AI/ML:** PyTorch, TensorFlow, Hugging Face, LLM Fine-tuning, NLP, Transformer Architectures
- Databases:** MongoDB, PostgreSQL, MySQL
- Tools & DevOps:** Git, GitHub, Shell Scripting, Linux, Docker, Grafana, Prometheus, Postman

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Operating Systems
- Computer Networks
- Object-Oriented Programming
- Database Management Systems
- Data Science